SET 4: THE ADIRI ICE-RIDGE DEUTERIUM RECOVERY & CRYOGENIC HYDROGEN STORAGE DEPOT

DOCUMENT 4A: SECURED CREDIT AGREEMENT

SECURED TERM CREDIT AGREEMENT

Dated as of the Agreement Effective Date (as defined herein)

among

ADIRI FUEL RESERVES LP,
as Borrower

and

SATURN ORBITAL DEVELOPMENT SYNDICATE (SODS),
as Administrative Agent, Collateral Agent, and Lead Arranger

and

THE FINANCIAL INSTITUTIONS PARTY HERETO FROM TIME TO TIME, as Lenders

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This SECURED TERM CREDIT AGREEMENT (this "Agreement") is entered into on the Agreement Effective Date by and among the Borrower, the Lenders, and the Administrative Agent. In consideration of the mutual covenants and agreements herein contained, the parties hereto agree as follows:

ARTICLE I: DEFINITIONS AND INTERPRETATION

Section 1.01. Defined Terms. As used in this Agreement, the following terms have the meanings specified below:

- **"Adiri Depot Structure"** means the physical, pressurized, and thermally insulated deuterium enrichment towers, centrifugal separators, and sub-surface liquid hydrogen containment tanks situated at Sector 12, Adiri Ice-Ridge coordinates (latitude 10.4° S, longitude 210.8° W) on Titan, Saturn.

- **"Agreement Effective Date"** means the date on which this Agreement has been fully executed and delivered by all parties hereto, as verified by the date inscribed next to the signatures of the authorized officers in the execution block below (such date, the **"Signature Date"**).

- **"Cryo-Storage Pressure Tolerance Index"** or **"CSPTI"** means the structural safety and pressure stability rating of the vacuum-jacketed, magnetic-suspension liquid hydrogen tanks, as evaluated on a

100-point index by the Titan Technical Safety Directorate (TTSD).

- **"Deuterium Extraction Yield Target"** means the mandatory minimum operational recovery metric of 1,200 kilograms of high-purity deuterium (D_2O equivalent) per Titan sidereal day, as calculated under the provisions of Section 4.01(2) herein.

- **"Helium-Boiloff Sinking Reserve"** means the liquid helium backup thermal ballast fund mandated to be maintained under Section 4.01(3) for secondary containment tank cooling.

- **"Sol System Interstellar Accounting Standards"** or **"SSIAS"** means the unified framework of financial accounting principles approved by the Interstellar Financial Board.

- **"TIDOR"** means the Titan Inter-Dome Offered Rate, which is the floating interbank benchmark rate calculated for Titan-denominated credit lines.

Section 1.02. Rules of Interpretation. Capitalized terms defined in the singular shall have the corresponding meanings in the plural, and vice versa. Any references to specific sections, tables, or documents relate directly to this Agreement unless otherwise specified.

ARTICLE II: THE SECURED CREDIT LINES

Section 2.01. The Facility. Subject to the terms and conditions of this Agreement, each Lender agrees to make a single advance term loan (each a "Facility Loan") to the Borrower on the **Closing Date** (which is identical to the **Signature Date**) in an aggregate principal amount of **4,200,000,000 Sol Credits (SC)**.

Section 2.02. Interest Rate Adjustments.

1. The Facility Loans shall accrue interest daily at a floating rate per annum equal to **TIDOR plus 5.75%**.
2. Should the **Cryo-Storage Pressure Tolerance Index (CSPTI)** fall below 92.0% for any consecutive 3-day cycle, the floating margin shall automatically increase by **2.50% per annum** (the "Pressure Containment Risk Premium") until the CSPTI is restored to 95.0% or above, as verified by the TTSD.

Section 2.03. Repayment and Amortization. Repayment of the principal shall be executed in equal quarterly installments on the final operational day of each Martian fiscal quarter, starting on the first such date following the **Agreement Effective Date**, in accordance with the amortization schedules appended hereto.

ARTICLE III: REPRESENTATIONS AND WARRANTIES

The Borrower represents and warrants to the Lenders that:

Section 3.01. Subsurface Water-Ice Mining Rights. The Borrower holds exclusive licenses to mine the subsurface water-ice regolith within Sector 12-Adiri, which serves as the primary hydrogen and deuterium feedstock under the authority of the Titan Regulatory Council (TRC).

Section 3.02. Passive Shielding and Cryo-Preservation. The liquid hydrogen storage chambers are insulated using a minimum of 40 layers of vacuum-reflective multi-layer insulation (MLI) and are anchored into thermal-isolation pylons engineered to prevent the conduct of environmental cold-sink deviations above 22.0 Kelvin.

ARTICLE IV: COVENANTS

Section 4.01. Affirmative Covenants. So long as any part of the Facility Loan remains outstanding, the Borrower shall:

1. **Storage System Maintenance:** Undertake continuous structural-imaging diagnostics of the primary liquid hydrogen containment vessels to ensure the CSPTI does not fall below the base standard of 93.0%.
2. **Maintain Deuterium Extraction Yield:** Operate the centrifugal separating columns to sustain the **Deuterium Extraction Yield Target** of 1,200 kilograms per Titan day, ensuring supply covenants with orbital fusion transit ships are satisfied.
3. **Thermal Ballast Escrow:** Maintain the cash-funded **Helium-Boiloff Sinking Reserve** with the Administrative Agent, holding a balance of not less than SC 300,000,000 at all times.

Section 4.02. Negative Covenants. The Borrower shall not, without the explicit written approval of the Administrative Agent:

1. **Containment Tank Alterations:** Modify, depressurize, or reroute the coolant pipelines of any active secondary storage tanks within the designated coordinates.

2. **Restricted Payments:** Declare or make any dividend distributions to corporate parent entities if the CSPTI is below 94.0% or if a Default Event has occurred and is continuing.

ARTICLE V: DEFAULT EVENTS AND REMEDIES

Section 5.01. Default Events. The occurrence of any of the following events shall constitute an Event of Default:

1. **Cryogenic Containment Failure:** Any sudden structural failure or loss of vacuum containment resulting in a rapid gas venting event exceeding 1,000 metric tons of gaseous hydrogen within any 24-hour cycle.

2. **Covenant Non-Compliance:** Failure to maintain the required balance in the **Helium-Boiloff Sinking Reserve** under Section 4.01(3), or failure to meet the **Deuterium Extraction Yield Target** for more than two consecutive quarters.

3. **Pylon Liquefaction or Anchoring Failure:** Any localized sub-surface heating that causes structural bedrock water-ice adjacent to the pylons to exceed 250 Kelvin, threatening the alignment or physical structural integrity of the fuel depot gantry.

ADIRI FUEL RESERVES LP

By: _____
Name: Julian Vance-Cross
Title: Chief Operating Officer, Fuel Reserves

Date of Execution: **June 26, 2146**

SATURN ORBITAL DEVELOPMENT SYNDICATE

By: _____
Name: Catherine Sterling
Title: Managing Partner, Outer-Rim Assets Division

Date of Execution (the **Signature Date**): **June 26, 2146**